

NOVA: QUERY-OCCURRENCE BINDING VIA LOCALLY VALID TIMESTAMP CONTRASTS FOR AUDIO TEMPORAL GROUNDING

Yuehan Zhang Zhicheng Tang Wei Xu*

Huazhong University of Science and Technology, Wuhan, China
yuehanzhang2005@hotmail.com, u202414252@hust.edu.cn

ABSTRACT

Audio temporal grounding must identify which occurrence answers a query when several acoustically valid windows coexist. Event presence alone does not determine which occurrence matches the requested relation. We introduce NOVA, combining relation-aware candidate generation, audio verification, and selective boundary refinement. Its core training module, RelTwin, learns query–occurrence binding through locally valid timestamp contrasts: shared excerpts form opposite-order windows whose correct and incorrect roles reverse with the query. Competing timestamp answers therefore differ in the requested relation, not acoustic validity. RelTwin reaches 59.43 mIoU on SpotSound-Bench and improves over matched SFT by 1.02 points, while joint localization improves by 31.25 points on synthetic relation-development data. NOVA reaches 86.86 mIoU on Clotho-Moment and 73.46 on the released UnAV subset. The binding advantage persists under timing intervention, supporting explicit supervision of which occurrence answers the query.

Index Terms— Audio temporal grounding, timestamp contrasts, query–occurrence binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query–occurrence binding error*: the wrong local occurrence is selected.

We propose NOVA, combining candidate learning for occurrence selection, audio verification for acoustic support, and refinement for boundary alignment. Our core training contribution is RelTwin (Fig. 1), which introduces *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows with real acoustic support; the query determines which window answers it. Sequence supervision

rewards correct answers individually, while candidate supervision compares both answers under each query. The trained generator directly outputs timestamps.

We formulate query–occurrence binding as a choice between locally valid timestamp answers. NOVA is evaluated on SpotSound, Clotho-Moment, and UnAV; matched comparisons and inverse-query tests show improved occurrence selection, which persists under timing intervention.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions [3]. Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned generation. TimeAudio adds temporal markers and absolute time-aware encoding [7]. SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8], combining presence supervision with timestamp prediction. RelTwin targets the query-specific choice between occurrences when inverse relations both have acoustic support.

Compositional audio–language learning. CLAP aligns audio and text representations [9]; T-CLAP adds order-reversed descriptions and temporal-focused contrast [10]. CompA evaluates event order and attribute binding, while CompA-CLAP trains with composition-aware negatives and modular contrast [11]. CoSTALA combines spatio-temporal contrast, local event–caption alignment, and feature consistency [12]. These methods organize compositional audio–text representations. RelTwin instead compares the query-conditioned likelihoods of complete timestamp answers, supervising selection in a grounding model’s output space.

Hard negatives for temporal grounding. For compositional video grounding, SHINE combines generated hard-negative queries with within-video and cross-query saliency ranking [13], comparing segment relevance and query compositions. RelTwin’s candidates are complete timestamp answers within one recording. Each denotes a real event sequence; reversing the query exchanges their labels while preserving the audio. Sequence likelihoods train this query-specific choice.

*Corresponding author: Wei Xu. Email: [to be provided].

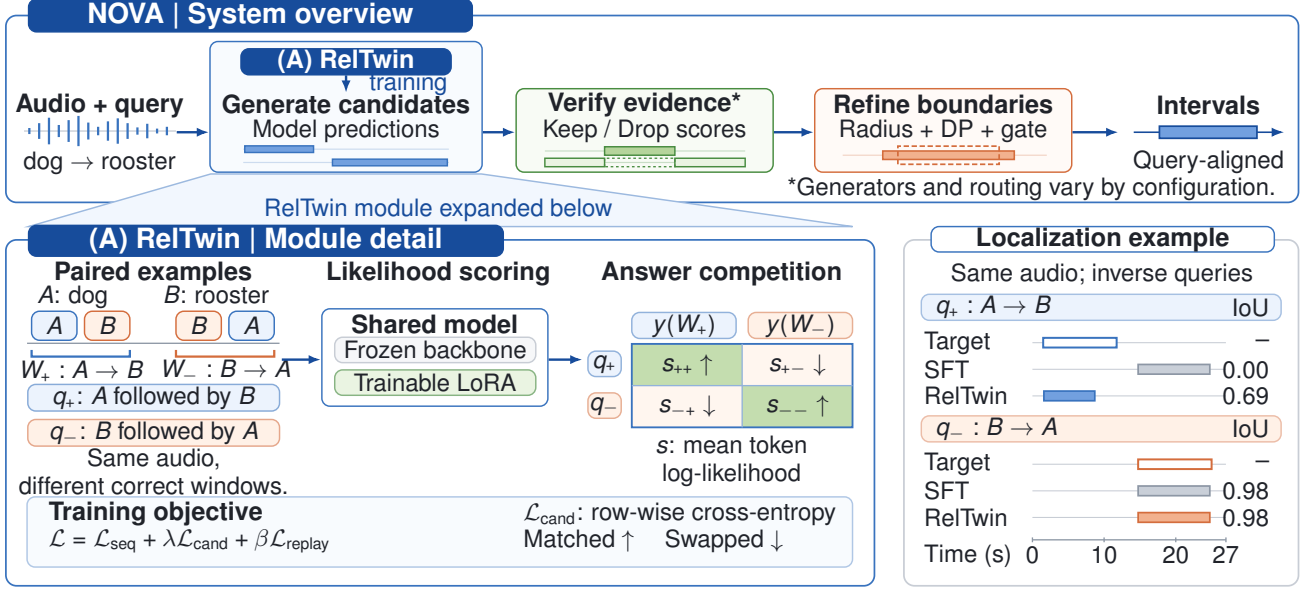


Fig. 1. NOVA with RelTwin’s training module (A) expanded below. Top: candidate generation, verification, and refinement (Section 4.1). Diagonal cells mark correct query–answer pairs. Right: on one relation-development recording, SFT repeats a window while RelTwin selects query-specific windows. Outlines: targets; filled bars: predictions; scores: set-IoU.

3. RELTWIN TIMESTAMP CONTRASTS

3.1. Two true relations, two different answers

RelTwin reuses ESC-50 excerpts [14] A and B to construct $A \rightarrow B$ and $B \rightarrow A$ in one recording x . Targets W_+ and W_- span from the first excerpt’s onset to the second’s offset, including the gap. Inverse queries q_+ and q_- request the respective orders, exchanging positive and negative labels for the same acoustically supported windows.

“Followed by” and “and then” provide two inverse-query pairs per recording; earlier-order layouts alternate (Section 5.3). Category rehearsal targets the union of both occurrences of A or B .

3.2. Contrast timestamp answers during training

We serialize an interval set S as comma-separated spans, each formatted like from 1.500s to 11.750s (three-decimal precision). Its supervised assistant-completion tokens $y(S)$ exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t \mid x, q, y_{<t}). \quad (1)$$

The 2×2 matrix $M_{ij} = s_\theta(x, q_i, W_j)$, $i, j \in \{+, -\}$, indexes queries by rows and answers by columns; diagonal labels $Y = I_2$ identify correct pairs (Fig. 1, module A). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, independently rewarding correct responses. RelTwin compares locally valid answers, training $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$ through row-wise probabilities $p_i(j) =$

$\text{softmax}_j(M_{ij}/\tau)$, $\tau > 0$. The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

For a separately sampled category example (x_r, q_r, S_r) , the rehearsal loss is $\mathcal{L}_{\text{replay}} = -s_\theta(x_r, q_r, S_r)$. Together,

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Paired SFT sets $\lambda = 0$; RelTwin uses $\lambda > 0$ with identical relation and rehearsal data. Both directly generate timestamps at inference, without candidates or an additional scorer.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ (zero for an empty union), and $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$ for predictions $\widehat{W}_i$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$; Swap error means $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc without Swap error. Scores are pair-level percentages; same-answer counts require identical parsed interval lists.

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails JointPairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Adaptation uses 40 ESC-50 classes: 256 recordings, 1,024 relation queries (512 pairs),

Table 1. Audio temporal grounding (%; higher is better). Baselines: SpotSound Table 3 [8]; NOVA configurations: Section 4.1. Bold marks column maxima; deltas use prior per-metric main-table bests. †NOVA uses the released 100-query UnAV subset; comparison protocols are in Section 4.

Method	SpotSound-Bench			Clotho-Moment			UnAV subset [†]		
	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5	mIoU	R1@.3	R1@.5
<i>Task-specific grounding models</i>									
WTATG	38.4	53.0	34.5	9.1	12.1	6.3	38.4	53.0	37.0
AM-DETR	19.5	25.5	15.0	80.9	89.8	88.0	42.8	59.0	46.0
<i>Proprietary audio–language models</i>									
Gemini-2.5-Flash	25.7	34.8	27.8	36.9	45.7	33.7	35.6	48.0	37.0
Gemini-2.5-Pro	19.7	20.0	17.3	32.5	40.4	32.5	34.6	39.0	35.0
<i>General-purpose open audio–language models</i>									
Kimi-Audio	3.2	3.0	1.3	0.9	0.7	0.1	5.3	8.0	2.0
Qwen2-Audio	4.4	5.5	1.8	5.7	6.1	1.3	9.7	14.0	4.0
Audio Flamingo 3	7.6	8.8	2.8	22.6	32.9	21.8	25.0	35.0	24.0
<i>Temporal grounding-enhanced models</i>									
TimeAudio	10.2	7.8	1.8	28.6	39.5	24.9	16.0	21.0	7.0
SpotSound-Q	49.9	65.0	51.3	85.4	93.6	91.2	72.4	88.0	74.0
SpotSound-A	57.9	76.5	59.5	85.6	93.4	91.0	69.8	86.0	74.0
NOVA	59.39	79.25	62.50	86.86	93.49	90.84	73.46	89.00	80.00
Prior main-table best	57.9	76.5	59.5	85.6	93.6	91.2	72.4	88.0	74.0
Δ vs. prior best	+1.49	+2.75	+3.00	+1.26	−0.11	−0.36	+1.06	+1.00	+6.00

and 512 rehearsal queries. Ten held-out classes provide 80 recordings, 320 queries, and 160 relation-development pairs, with disjoint original Freesound IDs. System evaluation uses 400 SpotSound queries, 6,649 Clotho-Moment queries, and the released 100-query/77-audio UnAV subset [2]. UnAV scores clip intervals to audio duration (unclipped mIoU: 73.39). SpotSound reports 997 UnAV queries; our evaluation uses the released 100-query subset. Development and public-benchmark scores were used during method development.

Training settings. SFT and RelTwin initialize from SpotSound-A on Audio Flamingo 3. LoRA [16] uses rank 8, $\alpha = 16$, dropout 0.1, and 20.19M parameters. Both run 256 AdamW [17] updates at learning rate 5×10^{-6} , zero weight decay, gradient clipping 1.0, $\tau = 1$, $\beta = 0.5$, and RelTwin $\lambda = 1$. Each update accumulates one inverse-query pair and one rehearsal query. The matched comparison shares initialization, data order, code, environment, and optimizer updates. RelTwin additionally computes four gradient-free candidate scores followed by weighted gradient recomputation, versus SFT’s two positive-answer backpropagations. Both include rehearsal.

Inference. SFT and RelTwin are evaluated after 256 updates. All generators use 16 kHz mono audio, bfloat16, and greedy one-beam decoding without sampling, capped at 128 new tokens. The post-development timing diagnostic uses the frozen matched checkpoints (Section 5.3).

Reference models. Table 2 includes paired SFT and the official backbone. The backbone scores 58.42 mIoU versus published 57.9; RelTwin’s three-seed difference against it is +0.75 points (audio-group CI $[-0.24, 1.74]$).

Result reporting and statistics. RelTwin point results use seed 0, selected by SpotSound mIoU from five runs, with its matched SFT control. Table 2(c) separately summarizes seeds 0/1/2 as mean $\pm$ sample SD. Training comparisons use 20,000 audio-group bootstrap resamples for 95% CIs. Relation comparisons resample 39 connected source-sharing groups (largest: 18 audios). CIs quantify audio-sample uncertainty conditional on the evaluated checkpoints; model selection is fixed across resamples.

4.1. NOVA system configurations

NOVA (Necessity-Oriented Verification over Audio) combines candidate generation, verification, and refinement. SpotSound uses RelTwin data with Jensen–Shannon consistency between p_+ and label-swapped p_- added to Eq. 3. Setwise preference optimization (SetPO) continues for 64 updates at 2×10^{-6} over six candidates: both windows, their expansions, enclosing span, and union. Soft targets combine set-IoU and interval-coverage quality. The RelTwin training comparison in Table 2 uses the three-term objective in Eq. 3.

Official, SFT, and SetPO seed-1 SpotSound predictions enter a LongNeedle-ESC50-fitted ridge router, using geometry and frozen query-existence scores after keeping/removing candidate intervals. Clotho and UnAV apply refinement to official predictions, whose initial mIoUs are 86.8543 and 73.4513. Refinement uses frozen SetPO/SpanTool features and ridge edit utilities: dynamic programming selects compatible boundaries, accepted when the 200-model bootstrap 5th percentile exceeds 0.02. Event count is preserved; radius

Table 2. SpotSound transfer and synthetic relation-development binding. (b) Matched data and updates; (c) mean $\pm$ sample SD (Section 4). Joint: JointPairAcc; same-answer: inverse-query pairs with identical timestamp answers. Scores are percentages except counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
RelTwin	59.43	88.55	84.38	7.50	1/160
RelTwin – Paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) RelTwin replication: three training seeds</i>					
RelTwin	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

$\min(0.25\text{ s}, 0.625\text{ event duration})$ is shared. AudioGrounding/AEGBench development evaluations selected the duration coefficient before these applications. Table 1 reports these NOVA configurations.

5. RESULTS AND ANALYSIS

5.1. Grounding performance across benchmarks

NOVA reaches 59.39 SpotSound mIoU (Table 1), exceeding the prior main-table best by 1.49 points. R1@.3 and R1@.5 improve by 2.75 and 3.00 points, respectively: gains extend from loose overlap to the stricter localization criterion. On Clotho-Moment, NOVA reaches 86.86 mIoU (+1.26), with R1@.3/.5 of 93.49/90.84. On UnAV, NOVA reaches 73.46 mIoU and 89.00/80.00 recall under the released-subset protocol in Section 4.

With direct timestamp generation, RelTwin reaches 59.43 SpotSound mIoU, improving over matched SFT by 1.02 points (CI [0.23, 1.81]). The comparison yields 92 wins, 239 ties, and 69 losses. The replication summary in Table 2(c) gives 59.16 mean mIoU. The following analyses examine how candidate supervision changes the response to inverse queries.

5.2. Timestamp contrasts recover query-specific selection

Paired SFT returns identical answers for 37/160 inverse-query pairs across 22 recordings; RelTwin reduces this to one (Table 2). Of those 37 pairs, 27 become jointly correct: the two predictions recover their respective requested relations. Figure 1 illustrates this query-specific recovery, with each inverse query selecting its corresponding event window.

JointPairAcc increases by 31.25 points (source-group CI [22.35, 40.50]) and Swap error decreases by 32.50. Across all pairs, 51 become jointly correct and one regresses. Relation mIoU increases by 13.97 points (CI [9.84, 17.51]), connecting selection with overlap.

Across three RelTwin seeds, JointPairAcc averages 81.46 and identical-answer counts are 1, 0, and 0, reproducing the reduction in same-answer collapse.

5.3. Binding gains persist after timing intervention

To test layout-specific silence, we recompose 80 development audios with 1.5 s lead-in, 0.25 s within-window gaps, 3 s between windows, and 2 s tail, preserving waveforms, order, and queries. The frozen matched models evaluate one template (160 queries). Timing changes for 40 BA-first recordings; 40 AB-first recordings remain waveform-identical.

On changed recordings, RelTwin reaches 87.50 JointPairAcc against SFT’s 67.50 (+20.00; CI [2.94, 33.33]); originally they scored 85.00 and 42.50. RelTwin retains a 20-point selection advantage under the altered timing, supporting query–occurrence binding across both silence layouts. Across all 160 queries, SFT and RelTwin score 81.34 and 90.65 mIoU.

5.4. Why locally valid answers matter

Shared excerpts hold event identity fixed while inverse queries exchange the correct window. Detecting both categories supports both candidates; selecting the requested order requires query-conditioned comparison. RelTwin’s reduced same-answer collapse and improved JointPairAcc capture this distinction in the controlled binding test. Timestamp-answer supervision preserves direct generation at inference.

6. CONCLUSION

NOVA learns query–occurrence binding with locally valid timestamp contrasts in RelTwin, complemented by verification and selective refinement. Matched comparisons and timing intervention demonstrate improved query-specific selection. Beyond detecting event presence, grounding benefits from learning which acoustically supported occurrence answers the query.

7. ACKNOWLEDGMENTS

Wei Xu personally funded GPU computing. No institutional or corporate funding was received. The authors declare no relevant financial or nonfinancial conflicts of interest.

Compliance with Ethical Standards. We reuse public ESC-50 [14], SpotSound [8], Clotho-Moment and UnAV [2], AudioGrounding [1], and AEGBench [18] data, including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

8. REFERENCES

- [1] Xuenan Xu, Heinrich Dinkel, Mengyue Wu, and Kai Yu, “Text-to-audio grounding: Building correspondence between captions and sound events,” *arXiv preprint arXiv:2102.11474*, 2021.
- [2] Hokuto Munakata, Taichi Nishimura, Shota Nakada, and Tatsuya Komatsu, “Language-based audio moment retrieval,” *arXiv preprint arXiv:2409.15672v3*, 2025.
- [3] Xuenan Xu, Ziyang Ma, Mengyue Wu, and Kai Yu, “Towards weakly supervised text-to-audio grounding,” *arXiv preprint arXiv:2401.02584*, 2024.
- [4] Soham Deshmukh, Benjamin Elizalde, Rita Singh, and Huaming Wang, “Pengi: An audio language model for audio tasks,” in *Advances in Neural Information Processing Systems*, 2023, vol. 36.
- [5] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, Chang Zhou, and Jingren Zhou, “Qwen2-Audio technical report,” *arXiv preprint arXiv:2407.10759*, 2024.
- [6] Arushi Goel, Sreyan Ghosh, Jaehyeon Kim, Sonal Kumar, Zhifeng Kong, Sang gil Lee, Chao-Han Huck Yang, Ramani Duraiswami, Dinesh Manocha, Rafael Valle, and Bryan Catanzaro, “Audio Flamingo 3: Advancing audio intelligence with fully open large audio language models,” *arXiv preprint arXiv:2507.08128*, 2025.
- [7] Hualei Wang, Yiming Li, Shuo Ma, Hong Liu, and Xiangdong Wang, “Listening between the frames: Bridging temporal gaps in large audio-language models,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, vol. 40, pp. 26233–26241.
- [8] Luoyi Sun, Xiao Zhou, Zeqian Li, Ya Zhang, Yanfeng Wang, and Weidi Xie, “SpotSound: Enhancing large audio-language models with fine-grained temporal grounding,” *arXiv preprint arXiv:2604.13023v2*, 2026.
- [9] Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Taylor Berg-Kirkpatrick, and Shlomo Dubnov, “Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation,” in *IEEE International Conference on Acoustics, Speech and Signal Processing*, 2023.
- [10] Yi Yuan, Zhuo Chen, Xubo Liu, Haohe Liu, Xuenan Xu, Dongya Jia, Yuanzhe Chen, Mark D. Plumbley, and Wenwu Wang, “T-CLAP: Temporal-enhanced contrastive language-audio pretraining,” *arXiv preprint arXiv:2404.17806*, 2024.
- [11] Sreyan Ghosh, Ashish Seth, Sonal Kumar, Utkarsh Tyagi, Chandra Kiran Reddy Evuru, Ramaneswaran S, S Sakshi, Oriol Nieto, Ramani Duraiswami, and Dinesh Manocha, “CompA: Addressing the gap in compositional reasoning in audio-language models,” in *International Conference on Learning Representations*, 2024.
- [12] Peiwei Ren, Jinbo Hu, Fang Kang, Shan Liang, and Yin Cao, “CoSTALA: Compositional spatio-temporal audio-language alignment via multi-grain hierarchical contrastive learning,” *arXiv preprint arXiv:2608.24374*, 2026.
- [13] Zixu Cheng, Yujiang Pu, Shaogang Gong, Parisa Kordjamshidi, and Yu Kong, “SHINE: Saliency-aware hierarchical negative ranking for compositional temporal grounding,” in *European Conference on Computer Vision*, 2024.
- [14] Karol J. Piczak, “ESC: Dataset for environmental sound classification,” in *Proceedings of the 23rd ACM International Conference on Multimedia*, 2015, pp. 1015–1018.
- [15] Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross, “Winoground: Probing vision and language models for visio-linguistic compositionality,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [16] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen, “LoRA: Low-rank adaptation of large language models,” in *International Conference on Learning Representations*, 2022.
- [17] Ilya Loshchilov and Frank Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations*, 2019.
- [18] Zihan Zhang, Xize Cheng, Wenhao Yan, Tong Zhang, Dongjie Fu, Boyun Zhang, Yongbo He, and Tao Jin, “Auto-AEG: Scalable data construction for open-vocabulary audio event grounding,” *arXiv preprint arXiv:2607.04383v4*, 2026.